

Efficacy of electroacupuncture combined with intravenous patient-controlled analgesia after cesarean delivery: a randomized clinical trial



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BACKGROUND: Electroacupuncture is a nonpharmacologic intervention for analgesia that is widely recognized as therapy for pain. However, the clinical efficacy of electroacupuncture combined with patient-controlled intravenous analgesia for postoperative analgesia after cesarean delivery remains unclear.

OBJECTIVE: This study aimed to assess the efficacy of electroacupuncture + patient-controlled intravenous analgesia for postoperative analgesia after cesarean delivery, determine the optimal frequency for the best analgesic effect, and explore the underlying mechanism of action.

STUDY DESIGN: This single-center, randomized, single-blinded, sham acupuncture controlled clinical trial was conducted at a tertiary university hospital in China. Female patients who underwent cesarean delivery and received fentanyl as patient-controlled intravenous analgesia for postoperative analgesia were enrolled. Patients were after surgery randomized to receive 2 Hz electroacupuncture treatment (n=53), 20/100 Hz electroacupuncture treatment (n=53), or sham electroacupuncture treatment (n=52) (controls). The 2 electroacupuncture groups received electroacupuncture treatment at 2 or 20/100 Hz at the ST36 and SP6 points, whereas, in the sham electroacupuncture group, sham electroacupuncture was performed at nonmeridian points with nonenergized electroacupuncture instruments. Of note, 4 electroacupuncture treatments were performed in all groups at 6, 12, 24, and 48 hours after surgery. The

primary outcome was the number of analgesic pump compressions at 48 hours after surgery. The secondary outcomes included number of analgesic pump compressions at 6, 12, and 24 hours after surgery; pain scores at 6, 12, 24, and 48 hours after surgery; fentanyl consumption at 48 hours after surgery; interleukin 6 and procalcitonin levels at 12 and 48 hours after surgery; and time to first exhaust.

RESULTS: Overall, 174 primigravida women were included in the intention-to-treat analysis. The number of analgesic pump compressions and pain scores at all 4 time points and fentanyl consumption at 48 hours after surgery were significantly lower in the electroacupuncture treatment groups than in the sham electroacupuncture group ($P < .001$).

CONCLUSION: Electroacupuncture + patient-controlled intravenous analgesia had a significantly better analgesic effect than sham electroacupuncture + patient-controlled intravenous analgesia within 48 hours after surgery. Thus, electroacupuncture can be considered safe and effective and may improve the efficacy of patient-controlled intravenous analgesia for pain management after cesarean delivery. Electroacupuncture can be recommended as a routine complementary therapy for pain control after cesarean delivery.

Keywords: analgesic pump compressions, cesarean delivery, electroacupuncture, fentanyl, pain scores, patient-controlled intravenous analgesia

Introduction

The overall rates of cesarean delivery (CD) have been increasing for the past 25 years.¹ Postoperative pain is one of the most common complications of CD, accounting for 54% of problems in postpartum women.^{2,3} Nearly 20% of women undergoing CD experience severe acute pain,⁴ which not only is an important predictor of chronic postoperative pain⁴⁻⁶ but also increases the probability of postpartum depression

and affects breastfeeding and infant health.³⁻⁹ As post-CD pain can seriously affect the physical and mental health of postpartum women, identification of the best analgesic regimen for its management is imperative (eg, postoperative multimodal analgesia).⁹⁻¹¹

To date, the most commonly used pain management method in clinical practice is patient-controlled epidural analgesia (PCEA) or patient-controlled intravenous analgesia (PCIA) with local anesthetics (such as ropivacaine) and opioids (such as fentanyl).^{4,12} Local anesthetics block the afferent transmission of nerve impulses in the dorsal root ganglion, whereas opioids act on opioid receptors in glial cells in the brain or spinal cord.¹³⁻¹⁵ When combined, they block the formation and conduction of pain information and increase the pain threshold to achieve analgesia.¹⁵ Although PCIA with

fentanyl can provide an excellent analgesic effect, the side effects of fentanyl, such as tendency to sedate and inhibit respiration, affecting the maternal heart rate and blood pressure, can lead to nausea and vomiting, hypotension, urinary retention, skin itching, and even low saturation.^{4,16,17} Therefore, although it is a conventional analgesic regimen, PCIA may exert potential safety hazards.

Acupuncture is a nonpharmacologic intervention for analgesia that has a significant effect on acute and chronic pain.¹⁸⁻²⁰ It has the advantages of easy operation, safety, and no toxic side effects and has been widely used in the treatment of postoperative pain in China and Western countries.^{21,22} The early postoperative analgesic effect of acupuncture is the key to successful prevention and reduction of chronic pain and related complications after CD.^{23,24}

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AJOG MFM at a Glance

Why was this study conducted?

This study aimed to investigate whether combining electroacupuncture (EA) with patient-controlled intravenous analgesia (PCIA) is an effective analgesic regimen after cesarean delivery (CD) and, if so, at which frequency and by what mechanism.

Key findings

In this randomized clinical trial, including 158 female patients, electroacupuncture + PCIA had significantly better clinical efficacy for reducing pain after CD than sham EA + PCIA, as shown by a reduced number of analgesic pump compressions, reduced pain scores, and decreased fentanyl consumption.

What does this add to what is known?

The findings from this clinical trial indicated that EA may improve the efficacy of PCIA for pain management after CD.

Some studies have demonstrated that electroacupuncture (EA) alone or combined with conventional analgesic regimens may be more effective than sham acupuncture (simulated acupuncture or acupuncture at nonmeridian points) in reducing pain scores and improving efficacy grades.^{25,26} However, a previous study found no significant difference in the clinical efficacy of acupuncture for postoperative analgesia after CD compared with that of placebo or pharmacologic analgesia.²⁷ The inconsistent findings may be related to differences in interventions, follow-up periods, and study population selection. Nevertheless, the clinical efficacy of EA combined with PCIA and the optimal frequency of EA for analgesic intervention remain unclear.

Therefore, this study aimed to assess the efficacy of EA combined with PCIA for postoperative analgesia after CD, determine the optimal frequency for the best analgesic effect, and explore the underlying mechanisms of action.

Materials and Methods

Study design and population

This single-center, randomized, single-blinded, sham acupuncture controlled clinical trial was conducted from January 1, 2021, to December 31, 2021, at the Department of Obstetrics at our hospital. The protocol was carried out following the Declaration of Helsinki

and was approved by the Ethics Review Committee of our hospital (permission number: 2021-0373). All participants provided written informed consent before participation. We followed the Consolidated Standards of Reporting Trials reporting guideline and the Standards for Reporting Interventions in Clinical Trials of Acupuncture guideline²⁸ for the designing and reporting of this trial. The trial protocol is available in Supplement 1.

Randomization and blinding

Eligible participants were randomized to receive PCIA + 2-Hz EA, PCIA + 20/100-Hz EA (EA groups), or PCIA + sham EA treatment (sham EA group) with a 1:1:1 ratio after CD (6, 12, 24, and 48 hours after surgery). The randomization sequence was generated at our hospital, and participants were randomized using the sealed envelope method. Physicians with experience in acupuncture provided the EA or sham EA intervention. Participants, outcome assessors, data analysts, and statisticians were blinded to treatment allocation, and acupuncturists were not.

Interventions

The acupuncturists who performed the treatment were qualified physicians with >5 years of experience. To improve maternal compliance, the same physician completed all acupuncture

treatments for the same patients. Each patient received 4 25-minute treatments at 6, 12, 24, and 48 hours after surgery. Patients were allowed to take anti-inflammatory analgesics as rescue medication for unbearable pain, and their use was recorded in detail.

Jiachen Brand sterile disposable acupuncture needles with 0.30- × 40-mm (Wujiang Jiachen Acupuncture Devices Co, Ltd, Jiangsu, China) were used in EA groups, whereas pragmatic placebo needles (with a blunt tip) with 0.25- × 25-mm were used in the sham EA group. All groups used the HuaTuo Electric Acupuncture stimulator (SDZ-IIB; Suzhou Medical Supplies Factory Co, Ltd, Jiangsu, China) with the power indicator lights covered and the battery compartment sealed to assist in blinding and to prevent the patient from adjusting settings other than intensity.

The EA groups received acupuncture at bilateral Zusanli (ST36) and Sanyinjiao (SP6) (eFigure 2 in Supplement 1), which were both inserted vertically to a depth of 25 to 30 mm by needles. The acupuncturists twirled the needle handles back and forth to achieve the sensation of heaviness and tightness (known as de qi) at all acupoints. The frequency was set at 2-Hz continuous wave in the 2-Hz group and at 20/100 Hz (EA with alternating administration of 20 Hz and 100 Hz at 3-second intervals) in the 20/100-Hz group, both with an intensity of 0.1–5.0 mA, which was appropriate for patient tolerance.

The sham EA group received sham acupoints (20 mm aside from the ST36 and SP6 points) (eTables 1 and 2 in Supplement 1) using pragmatic placebo needles. Procedures, electrode placements, and other treatment settings were similar to those in the EA groups, but with no skin penetration, electricity output, or needle manipulation for de qi.

Outcome measures

The primary outcome was the change in the number of analgesic pump compressions (NAPC) at 48 hours after surgery. The secondary outcomes included pain scores, fentanyl consumption at 48 hours after surgery, serum levels of interleukin 6 (IL-6) and procalcitonin (PCT) at 12

and 48 hours after surgery, and time to first exhaust.

Any adverse event, such as unfavorable or unintended signs, symptoms, or diseases, related to the acupuncture treatment or the administration of fentanyl was reported by patients and the acupuncturists. Severe adverse events were reported to the principal investigator and the data and safety monitoring board within 24 hours after their occurrence.

Statistical analysis

Data analysis was performed from February 4, 2022, to June 13, 2022. Data are expressed as median (interquartile range [IQR]) or mean $\pm$ standard deviation (SD). The Shapiro-Wilk test combined histogram, P-P plot, and Q-Q plot were performed to confirm whether the data were normally distributed. Normally distributed measurement data are expressed as mean $\pm$ SD and were analyzed using the 1-way analysis of variance. Skewed variables are expressed as IQR and were analyzed using the Kruskal-Wallis test. If the overall significance test was significant, pairwise comparisons were performed using the least significant difference (LSD) test when the data were normally distributed; conversely, rank-transformed LSD was used when the

variables were nonnormally distributed. For IL-6 and PCT changes (12 and 48 hours after surgery), we planned to use the paired *t* test and paired rank-sum test for normally and nonnormally distributed data, respectively. All data were analyzed using IBM SPSS Statistics (version 25.0; IBM Corporation, Armonk, NY). Statistical significance was set at $P < .05$. Posthoc Bonferroni was used for multiple comparisons, and statistical significance was set at $P < .05/k$ (*k* for number of comparisons).

Results

Patients and baseline characteristics

A total of 453 patients were planned scheduled for CD, of which 174 eligible patients were enrolled; 250 patients were excluded for different reasons, and 29 were not randomized. The 174 patients were randomized to receive PCIA + 2-Hz EA ($n=58$), PCIA + 20/100-Hz EA ($n=58$), or PCIA + sham EA ($n=58$) treatment (Figure 1). Of note, 16 patients (9.2%) dropped out of the study owing to time commitment, personal issues, or unsatisfying effects (5 in the 2-Hz EA group, 5 in the 20/100-Hz EA group, and 6 in the sham EA group). Of the patients who completed the intervention, 53 were in the 2-Hz EA group, 53 were in the 20/100

-Hz EA group, and 52 were in the sham EA group. There was no significant difference in the baseline characteristics of patients among the groups ($P > .05$) (Table 1).

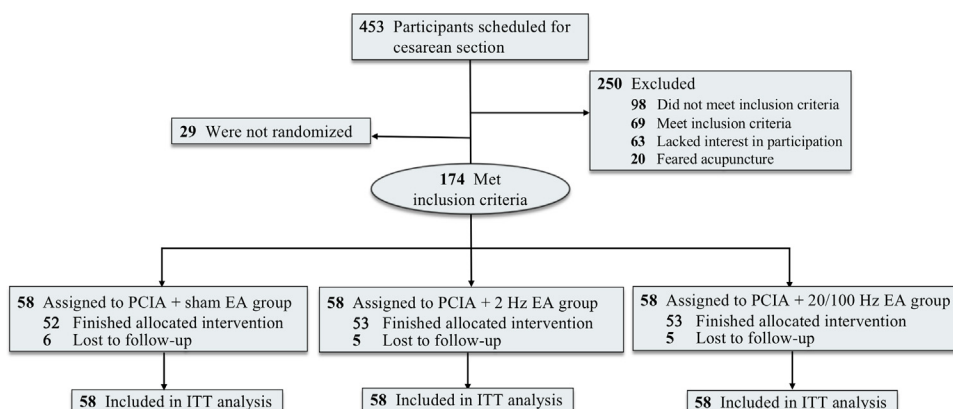
Primary outcome

Significant differences were observed in the NAPC among the 3 groups at 48 hours after surgery (48 hours: $H=63.68$; $P < .001$). After pairwise comparison, a significant difference in the NAPC was observed between the sham EA group and the 2-Hz EA ($P < .001$ after adjustment) and 20/100-Hz EA groups ($P < .05$ after adjustment), but not between the 2 EA groups ($P > .05$ after adjustment) (Table 2).

Secondary outcomes

There were differences in the number of compressions among the 3 groups at 6, 12, and 24 hours after surgery (6 hours: $H=25.49$; $P < .001$; 12 hours: $H=40.37$; $P < .001$; 24 hours: $H=64.50$; $P < .001$). After pairwise comparison, a significant difference in the NAPC was observed between the sham EA group and the 2-Hz EA ($P < .001$ after adjustment) and 20/100-Hz EA groups ($P < .05$ after adjustment), but not between the 2 EA groups ($P > .05$ after adjustment) (Table 2).

FIGURE 1
Study flowchart



EA, electroacupuncture; ITT, intention to treat; PCIA, patient-controlled intravenous analgesia.

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TABLE 1
Baseline characteristics of the intention-to-treat analysis

Characteristic	Group			F value	P value
	PCIA + sham EA group (n=52)	PCIA + 2-Hz EA group (n=53)	PCIA + 20/100-Hz EA group (n=53)		
Age (y)	31.19±3.37	30.83±3.59	30.13±3.31	1.30	.28
Height (cm)	159.44±5.51	159.87±4.86	159.70±5.38	.09	.92
Weight (kg)	67.80±9.38	69.37±9.11	71.15±11.37	1.47	.23
BMI (kg/m ²)	26.65±3.22	27.16±3.47	27.82±3.54	1.56	.21
Gestational (wk)	38.19±1.28	37.96±1.64	38.32±1.45	.82	.44

Data are presented as mean±standard deviation.

BMI, body mass index; EA, electroacupuncture; PCIA, patient-controlled intravenous analgesia; SD, standard deviation.

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The effect of EA on the secondary outcomes seemed to persist throughout the study (Table 3). There was no statistical difference in the pain scores among the 3 groups at 6 hours after surgery ($P>.05$); however, pain scores at 12, 24, and 48 hours after surgery were significantly lower in the two EA groups than in the sham EA group, with little difference between the 2-Hz and 20/100-Hz groups ($P>.05$). The decrease in pain scores over time in the sham EA group was significantly smaller than that in the EA groups (Figure 2).

Fentanyl consumption was significantly lower in the 2 EA groups than in the sham EA group at all time points ($P<.05$), with a significant difference between the 2-Hz and 20/100-Hz groups only at 24 hours after surgery (difference, 6.41; 95% confidence interval [CI], 0.61–12.21; $P=.0031$) (Table 3).

The analysis of IL-6 and PCT indicators revealed no significant differences in their levels among the 3 groups at 12 and 48 hours after surgery ($P>.05$), and a paired rank-sum test showed that the paired test of IL-6 and PCT level changes (12 and 48 hours after surgery) in the blood of the 3 groups at 12 and 48 hours after surgery were statistically significant ($P<.05$), suggesting that the improvement of symptoms was related to the changes in IL-6 and PCT content. Regarding the IL-6 index, the paired test of IL-6 levels at 12 and 48 hours after surgery in each group showed a statistically significant difference

($P<.05$), and no significant difference was observed among the 3 groups. In terms of PCT levels, the paired test results of PCT content at 12 hours after surgery and 48 hours after surgery in the sham EA group and 20/100-Hz EA group were significantly different ($P<.05$), and even the paired test results of PCT content at 12 hours after surgery and 48 hours after surgery in the 2-Hz EA group were not significantly different ($P=.208$), which required verification using an accurate test with a large sample size (Table 4).

There was no significant difference in the time to first exhaust between the 2 EA groups and the sham EA group ($P>.05$) (Table 3).

Safety

Overall, 5 patients reported adverse events. Of note, 1 patient (1.89%) from the 2-Hz EA group reported skin bruising at ST36, 3 patients (5.66%) from the 20/100-Hz EA group experienced subcutaneous hematoma at the puncture site, and 1 patient (1.92%) from the sham EA group developed a subcutaneous hematoma in the burrow area. No significant difference in the proportion of patients with adverse events was observed among the groups ($P=.10$). No patient had severe adverse events in the trial (eTable 1 in Supplement 2). All patients recovered from the adverse events and were not withdrawn from the trial.

Discussion

Principal findings

Our study showed that PCIA + EA had significantly better efficacy than PCIA + sham EA in reducing the NAPC, pain scores, and fentanyl consumption at 12, 24, and 48 hours after CD. However, IL-6 and PCT levels at 12 and 48 hours after surgery were not significantly different among the 3 groups. No patient reported severe adverse events in the trial. These findings indicated that EA may improve the efficacy of PCIA for pain management after CD.

Results and clinical implications

Compared with the sham EA group, the 2 EA groups showed significant advantages in various outcome measures at 12, 24, and 48 hours after surgery. However, there was no significant difference in the pain scores among the 3 groups at 6 hours after surgery, which may indicate that anesthesia affected pain assessment within 6 hours.

According to Duan et al,²⁹ visceral pain after CD mainly occurs in the early postoperative period; therefore, we speculated that early analgesia after CD may be based on the inhibition of visceral pain. Post-CD visceral pain is mainly due to uterine contraction, leading to ischemia and hypoxia at the uterine base, which together with the noxious stimulation brought about by the surgical incision, jointly stimulates the release of endogenous analgesic substances in the uterine tissue. These

TABLE 2
Number of compressions of the analgesic pump after surgery

Outcomes	Comparison among groups					
	PCIA + sham EA vs PCIA + 2-Hz EA		PCIA + sham EA vs PCIA + 20/100-Hz EA		PCIA + 2-Hz EA vs PCIA + 20/100-Hz EA	
	P value ^a	Difference (95% CI)	P value ^b	Difference (95% CI)	P value ^b	Difference (95% CI)
6 h after surgery	<.001	1.50 (0.73–2.29)	<.001	1.20 (0.44–1.97)	.001	–0.30 (–0.78 to 0.18)
12 h after surgery	<.001	3.58 (2.28–4.87)	<.001	3.48 (2.21–4.75)	<.001	–0.09 (–0.78 to 0.59)
24 h after surgery	<.001	5.72 (4.10–7.34)	<.001	5.57 (3.98–7.15)	<.001	–0.15 (–1.09 to 0.78)
48 h after surgery	<.001	7.91 (5.81–10.00)	<.001	7.04 (4.84–9.24)	<.001	–0.87 (–2.27 to 0.54)

Data are presented as median (interquartile range).
 CI, confidence interval; EA, electroacupuncture; PCIA, patient-controlled intravenous analgesia; NA, not applicable.
^a P value based on Kruskal-Wallis analysis among the 3 groups (adjusted value); ^b P value based on the analysis of variance and least significant difference analysis among the 3 groups.
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endogenous substances, in turn, activate the corresponding receptors, inducing transmission of the stimuli to the dorsal spinal roots at T10 to T12 and L1 levels through the A and C pain impulse conduction fibers of the pelvic visceral nerves.³⁰ It has been shown that EA has a unique role in inhibiting the release of endogenous analgesic substances³¹ and can effectively inhibit inflammatory mediators, such as 5-hydroxytryptamine, nerve growth factor, tumor necrosis factor alpha, and prostaglandin E2,³² which can block the transmission of noxious stimuli, increase the pain threshold, and effectively reduce the pain after CD. In addition, acupuncture at ST36 was found to promote glucose metabolism in the dorsal middle pons and midbrain and activate large nuclear neurons in the brainstem raphe, thereby inhibiting nociceptive responses, such as pain,³³ whereas acupuncture at SP6 shows unique superiority in relieving uterine contraction pain by reducing the spasm of uterine smooth musculature, relieving the uterine microcirculation disturbance and local tissue ischemia and hypoxia, and regulating the oxidative stress response of the uterine tissue.³⁴ Therefore, targeted selection of EA stimulation at ST36 and SP6 points can effectively block the transmission of pain information, increase the pain threshold, and inhibit pain formation. Wang et al^{35,36} reported that EA stimulation at SP6 can effectively delay the time to analgesia request after CD and reduce the dose of analgesic drugs used within 24 hours. These findings were consistent with the results of our study.

Fentanyl consumption at 24 hours after surgery was significantly lower in the 2-Hz EA group than in the 20/100-Hz EA group, suggesting that 2-Hz EA combined with PCIA may exert a more considerable analgesic effect within 24 hours after CD. Previous studies have demonstrated that 2-Hz EA can stimulate the central nervous system to release endogenous opioid analgesic substances, such as enkephalin and endorphin, which act on μ and δ receptors, respectively, and activate the periaqueductal gray matter of the

TABLE 3

Pain score, fentanyl consumption, and the time of the first exhaust after surgery

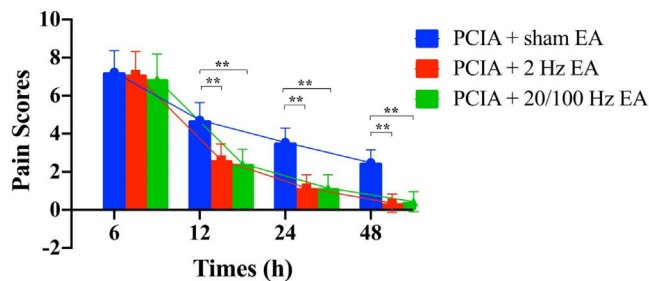
Outcomes	PCIA + sham EA group (n=52)	PCIA + 2-Hz EA group (n=53)	PCIA + 20/100—Hz EA group (n=53)	Statistic	<i>P</i> value ^a	Comparison among groups					
						PCIA + sham EA vs PCIA + 2-Hz EA	<i>P</i> value ^b	PCIA + sham EA vs PCIA + 20/100—Hz EA	<i>P</i> value ^b	PCIA + 2-Hz EA vs PCIA + 20/100—Hz EA	<i>P</i> value ^b
						Difference (95% CI)		Difference (95% CI)		Difference (95% CI)	
Pain score 6, 12, 24, and 48 h after surgery, mean±SD ^b											
6 h after surgery	7.23±1.13	7.17±1.21	6.87±1.32	1.34	.266	0.06 (−0.41 to 0.53)	.798	0.36 (−0.11 to 0.83)	.130	0.30 (−0.17 to 0.77)	.205
12 h after surgery	4.71±0.92	2.62±0.84	2.42±0.77	118.98	<.001	2.09 (1.76–2.41)	<.001	2.30 (1.97–2.62)	<.001	0.21 (−0.12 to 0.53)	.207
24 h after surgery	3.54±0.75	1.15±0.69	1.15±0.69	196.31	<.001	2.39 (2.11–2.66)	<.001	2.39 (2.11–2.66)	<.001	0.00 (−0.27 to 0.27)	1.000
48 h after surgery	2.48±0.67	0.62±0.48	0.58±0.49	197.27	<.001	1.86 (1.64–2.07)	<.001	1.90 (1.68–2.11)	<.001	0.04 (−0.18 to 0.25)	.728
Fentanyl consumption 6, 12, 24, and 48 h after surgery, mean±SD ^b											
6 h after surgery (mL)	14.61±6.55	10.16±5.21	11.49±4.21	9.37	<.001	4.45 (2.37–6.53)	<.001	3.12 (1.04–5.20)	.004	−1.33 (−3.40 to 0.74)	.207
12 h after surgery (mL)	30.29±10.47	20.32±8.69	22.09±8.32	17.50	<.001	9.97 (6.42–13.52)	<.001	8.20 (4.65–11.75)	<.001	−1.77 (−5.30 to 1.76)	.324
24 h after surgery (mL)	56.54±12.50	39.31±15.58	45.72±16.92	17.38	<.001	17.24 (11.40–23.07)	<.001	10.83 (5.00–16.66)	.001	−6.41 (−12.21 to 0.61)	.031
48 h after surgery (mL)	94.88±9.91	82.41±21.18	83.76±17.91	8.45	<.001	12.47 (5.90–19.04)	<.001	11.12 (4.55–17.69)	.001	−1.35 (−7.89 to 5.19)	.684
The time of the first exhaust after surgery, mean±SD ^b											
The first exhaust after surgery (h)	32.19±12.93	30.77±13.10	29.88±14.07	.40	.675	1.41 (−3.75 to 6.58)	.589	2.30 (−2.86 to 7.46)	.380	1.41 (−3.75 to 6.57)	.589

CI, confidence interval; EA, electroacupuncture; PCIA, patient-controlled intravenous analgesia; NA, not applicable; SD, standard deviation.

^a Bonferroni method corrected, the P value of <.0167 indicated that it was statistically significant different; ^b P value based on Kruskal-Wallis analysis among the three groups (adjusted value).

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FIGURE 2
Pain score curves at 6, 12, 24, and 48 hours after surgery



EA, electroacupuncture; PCIA, patient-controlled intravenous analgesia.

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midbrain, thereby inhibiting the generation of pain perception in the parathalamic nucleus through ascending analgesia and inhibiting the ascending pain signal transduction pathway along the posterior horn cells, resulting in a slow and long-lasting analgesic effect.^{37,38} To date, 2-Hz EA is rarely applied for postoperative analgesia after CD, which may be related to the degree of pain in patients (ie, personality differences).

Here, EA did not reduce the levels of IL-6 and PCT, which is inconsistent with previous findings^{39–42} that

acupuncture is effective in reducing the release of interleukin pain-inducing substances from immune cells. Other studies have also found that the effective application of multimodal analgesic regimens after CD is mostly accompanied by varying degrees of decrease in IL-6 levels.⁴³ In addition, in a basic experimental study, EA could reduce serum IL-6 levels in rats, thereby inhibiting related neuroinflammation and relieving pain symptoms.⁴⁴ Therefore, we still believe that there is a certain correlation between EA and IL-6 levels, which could not be observed in the current

study because of the small sample size. Few studies have explored the relationship between PCT and EA for postoperative analgesia after CD. Hence, based on the available evidence and the results of our study, we cannot speculate that the analgesic effect of EA in postoperative analgesia after CD is related to the release of IL-6 and PCT.

Research implications

There are still unanswered questions about the mechanism by which EA combined with PCIA may have a better clinical efficacy after CD. Further randomized controlled trial studies should be needed to determine whether EA combined with PCIA would be associated with reducing the release of interleukin pain-inducing substances from immune cells and other potential mechanisms for improvements and whether the results from experimental studies in rats is sufficient to support it.

Strengths and limitations

This was an intention-to-treat analysis on the efficacy of EA + PCIA for postoperative analgesia after CD to determine the optimal frequency for the best analgesic effect and explore the underlying

TABLE 4
Changes in IL-6 and PCT levels after surgery in the 3 groups

Index	Outcomes	PCIA + sham EA group (n=52)	PCIA + 2-Hz EA group (n=53)	PCIA + 20/100-Hz EA group (n=53)	Statistic	P value ^a
IL-6 content	12 h after surgery	60.69 (48.96–80.24)	68.41 (49.42–83.52)	58.18 (44.70–86.04)	1.07	.587
	48 h after surgery	17.19 (12.65–23.90)	15.92 (9.84–20.05)	15.16 (12.61–21.90)	3.21	.201
	Z value ^b	–6.21	–6.33	–6.31		
	P value ^b	<.001	<.001	<.001		
PCT content	12 h after surgery	0.11 (0.08–0.18)	0.11 (0.08–0.17)	0.11 (0.08–0.18)	0.48	.201
	48 h after surgery	0.08 (0.06–0.13)	0.09 (0.05–0.20)	0.07 (0.05–0.12)	2.75	.253
	Z value ^b	–3.25	–1.26	–4.08		
	P value ^b	.001	.208	<.001		

EA, electroacupuncture; IL-6, interleukin 6; PCIA, patient-controlled intravenous analgesia; PCT, procalcitonin.

^a P and Z values based on the Wilcoxon signed-rank test of IL-6 and PCT at 12 and 48 hours after surgery in the 3 groups; ^b P and Z values based on the Wilcoxon signed-rank test of IL-6 at 12 and 48 hours after surgery in each group.

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mechanism of action. Data came from a single-center, randomized, single-blinded, sham acupuncture controlled clinical trial that was conducted at a tertiary university hospital in China. With our study design, we were successful in recruiting a study population, and we lost very few patients from the study.

Our study has some limitations. First, the single-center nature of our study design relied on the expertise of the treating clinicians, limiting patient demographic variability. Second, follow-up records after 48 hours from CD were insufficient to predict the late analgesic efficacy of EA. Finally, our study could not exclude confounding factors other than treatment, such as mood and sleep. Future prospective large-scale studies are required to address these limitations.

Blinding

The blinding assessment results (eTable 2 in Supplement 2) showed that 15 patients (28.8%) in the sham EA group (Bang blinding index, -0.4 [95% CI, -0.6 to -0.3]), 11 patients (20.8%) in the 2-Hz EA group, and 12 patients (22.6%) in the 20/100-Hz group made a wrong guess regarding their group assignment. No statistically significant difference in the blinding index was observed among the 3 groups ($P > .05$). However, the results indicated a relatively higher degree of blinding in the control group.

Conclusions

This study found that EA combined with PCIA had a significantly better clinical efficacy after CD. Given these results, EA is recommended as a routine complementary therapy for pain control after CD and may improve the efficacy of PCIA for pain management after CD. ■

Supplementary materials

Supplementary material associated with this article can be found in the online version at [doi:10.1016/j.ajogmf.2022.100826](https://doi.org/10.1016/j.ajogmf.2022.100826).

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